

Deeksha Sinha

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Education

Northeastern University

M.S. in Computer Science

Boston, MA

Jan 2025 – **Fall 2026 (Expected)**

GPA: 4.0/4.0 — Data Structures & Algorithms, Distributed Systems, Machine Learning **West Bengal Technical**

University

B.Tech in Computer Science and Engineering

Kolkata, India

2013 – 2017

Experience

Amazon Web Services (AWS)

Arlington, VA

Summer Intern

May 2026 – Aug 2026

- Built **DeployGuard**, a distributed pre-deployment validation service on EKS Fargate that validates ECS configurations against live AWS resources, preventing **95% of identified deployment misconfigurations** before production.
- Engineered secure, low-latency infrastructure using **Kubernetes, IRSA, NLB IP-mode networking, and VPC endpoints**, achieving **sub-100ms p99 latency** and provisioning infrastructure as code with AWS CDK.
- Built an automated deployment gate using **CodePipeline, CodeBuild, ECR, and Kubernetes rolling deployments**; integrated **Amazon Bedrock** to generate remediation suggestions and resolved **11 security findings** with zero critical vulnerabilities remaining.

J.P. Morgan Chase & Co.

Bangalore, India

Software Development Engineer

Sep 2023 – Jul 2024

- Designed Spring Boot and AWS **distributed microservices** processing **100K+ transactions/day** at **99.9% uptime**, using circuit breakers, exponential-backoff retries, graceful degradation, and idempotency.
- Built event-driven ingestion pipelines using **AWS Lambda, Glue, and S3** with asynchronous processing and eventual consistency, reducing operational costs by **90%**.
- Developed secure **OAuth 2.0 REST APIs** with rate limiting and idempotency guarantees, integrating with **20+ internal services**.

Ninjacart

Bangalore, India

Software Development Engineer

Jul 2022 – Sep 2023

- Architected a BNPL platform using Kafka partitioned topics, consumer groups, DLQs, and exactly-once processing semantics, reducing external payment dependency by **60%**.
- Built **GEMS**, a schema-driven REST API generation platform using Apicurio Registry and MongoDB for schema evolution, reducing organization-wide API development time by **80%**.
- Implemented **Grafana, Prometheus, distributed tracing, and SLO-based alerting**, reducing mean time to recovery (MTTR) by **50%**.

Manhattan Associates

Bangalore, India

Software Development Engineer, R&D

Jan 2020 – Jul 2022

- Built a distributed **forecasting engine** using parallel computation across partitioned datasets; automated CI/CD with **blue/green deployments** and rollback automation.

Capgemini Technology Services

Bangalore, India

Software Development Engineer

Oct 2017 – Jan 2020

- Led a **monolith-to-microservices migration** using Spring Boot, Docker, and the Strangler Fig pattern, reducing release cycles by **60%**; also built an NLP classification engine processing 10K+ messages/day.

Technical Skills

Languages: Python, Java, Go, C++, JavaScript, SQL, Bash

Backend & Distributed Systems: Spring Boot, Microservices, Kafka, RabbitMQ, REST APIs, Event-Driven Architecture, CQRS, Saga, Idempotency

Cloud & Infrastructure: AWS, Kubernetes, Docker, Terraform, AWS CDK, CloudFormation, CI/CD, EKS, ECS, Lambda

Data & Observability: DynamoDB, PostgreSQL, MongoDB, Redis, Elasticsearch, Grafana, Prometheus, CloudWatch

AI/ML: Amazon Bedrock, LangChain, Transformers, scikit-learn, Generative AI, Claude, Agentic Systems

Selected Projects

Skier Tracking System — Built a high-throughput distributed backend processing **200K+ requests/day** using Spring Boot, EC2, RabbitMQ, and DynamoDB with rate limiting and back-pressure.

Bundle Recommendation — Developed a Graph Transformer using self-supervised learning and causal reasoning, improving **Recall@K** by **12.8%** and **NDCG@K** by **10.5%**.

Hirely.AI — Built a distributed GenAI backend with multi-agent orchestration, embedding pipelines, and asynchronous processing, reducing screening time by **70%**.

Certifications & Leadership

AWS Solutions Architect Professional (SAP-C02) — **AWS SA Associate (SAA-C03)** — **AWS AI**

Practitioner (AIF-C01) — **AWS Cloud Practitioner (CLF-C02)**

Oracle Certified Java Professional — Women Who Code – Mentor — AnitaB.org / GHC25 / RTC – Member